US Patent & Trademark Office
Patent Public Search | Text View

United States Patent	12409075
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Locke; Christopher Brian et al.

Dressing with contracting layer for linear tissue sites

Abstract

Systems, apparatuses, and methods for closing an opening through a surface of a tissue site are described. The system includes a sealing member adapted to cover the opening to form a sealed space and a negative-pressure source adapted to be fluidly coupled to the sealed space to provide negative pressure to the sealed space. The system also includes a contracting layer adapted to be positioned adjacent the opening and formed from a material having a firmness factor and a plurality of holes extending through the contracting layer to form a void space. The holes have a perforation shape factor and a strut angle causing the plurality of holes to collapse in a direction substantially perpendicular to the opening. The contracting layer generates a closing force substantially parallel to the surface of the tissue site to close the opening in response to application of the negative pressure.

Inventors: **Locke; Christopher Brian (Bournemouth, GB), Robinson; Timothy Mark (Shillingstone, GB), Tuck; Robert (Hampshire, GB), Stephenson; Gareth (Southampton, GB)**

Applicant: **Solventum Intellectual Properties Company** (Maplewood, MN)

Family ID: **53191845**

Assignee: **Solventum Intellectual Properties Company (Maplewood, MN)**

Appl. No.: **16/918682**

Filed: **July 01, 2020**

Prior Publication Data

Document Identifier	Publication Date
US 20200330284 A1	Oct. 22, 2020

Related U.S. Application Data

continuation parent-doc US 15958773 20180420 US 10736788 child-doc US 16918682
continuation parent-doc US 14708061 20150508 US 9974694 20180522 child-doc US 15958773
us-provisional-application US 61991174 20140509

Publication Classification

Int. Cl.: **A61F13/05** (20240101); **A61F13/01** (20240101); **A61M1/00** (20060101)

U.S. Cl.:

CPC **A61F13/05** (20240101); **A61F13/01017** (20240101); **A61F13/01042** (20240101); **A61M1/915** (20210501); A61M1/92 (20210501); A61M1/94 (20210501)

Field of Classification Search

CPC: A61F (13/00068); A61F (13/00017); A61F (13/00042); A61F (13/01017); A61F (13/01042); A61F (13/05); A61M (1/915); A61M (1/92); A61M (1/94)

USPC: 606/213

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
1355846	12/1919	Rannells	N/A	N/A
2547758	12/1950	Keeling	N/A	N/A
2632443	12/1952	Lesher	N/A	N/A
2682873	12/1953	Evans et al.	N/A	N/A
2910763	12/1958	Lauterbach	N/A	N/A
2969057	12/1960	Simmons	N/A	N/A
3066672	12/1961	Crosby, Jr. et al.	N/A	N/A
3367332	12/1967	Groves	N/A	N/A
3520300	12/1969	Flower, Jr.	N/A	N/A
3568675	12/1970	Harvey	N/A	N/A
3648692	12/1971	Wheeler	N/A	N/A
3682180	12/1971	McFarlane	N/A	N/A
3683921	12/1971	Brooks et al.	N/A	N/A
3826254	12/1973	Mellor	N/A	N/A
4080970	12/1977	Miller	N/A	N/A
4096853	12/1977	Weigand	N/A	N/A
4139004	12/1978	Gonzalez, Jr.	N/A	N/A
4165748	12/1978	Johnson	N/A	N/A
4184510	12/1979	Murry et al.	N/A	N/A
4233969	12/1979	Lock et al.	N/A	N/A
4245630	12/1980	Lloyd et al.	N/A	N/A
4256109	12/1980	Nichols	N/A	N/A
4261363	12/1980	Russo	N/A	N/A
4275721	12/1980	Olson	N/A	N/A
4284079	12/1980	Adair	N/A	N/A

4297995	12/1980	Golub	N/A	N/A
4333468	12/1981	Geist	N/A	N/A
4373519	12/1982	Errede et al.	N/A	N/A
4382441	12/1982	Svedman	N/A	N/A
4392853	12/1982	Muto	N/A	N/A
4392858	12/1982	George et al.	N/A	N/A
4419097	12/1982	Rowland	N/A	N/A
4465485	12/1983	Kashmer et al.	N/A	N/A
4475909	12/1983	Eisenberg	N/A	N/A
4480638	12/1983	Schmid	N/A	N/A
4525166	12/1984	Leclerc	N/A	N/A
4525374	12/1984	Vaillancourt	N/A	N/A
4540412	12/1984	Van Overloop	N/A	N/A
4543100	12/1984	Brodsky	N/A	N/A
4548202	12/1984	Duncan	N/A	N/A
4551139	12/1984	Plaas et al.	N/A	N/A
4569348	12/1985	Hasslinger	N/A	N/A
4605399	12/1985	Weston et al.	N/A	N/A
4608041	12/1985	Nielsen	N/A	N/A
4640688	12/1986	Hauser	N/A	N/A
4655754	12/1986	Richmond et al.	N/A	N/A
4664662	12/1986	Webster	N/A	N/A
4710165	12/1986	McNeil et al.	N/A	N/A
4733659	12/1987	Edenbaum et al.	N/A	N/A
4743232	12/1987	Kruger	N/A	N/A
4758220	12/1987	Sundblom et al.	N/A	N/A
4787888	12/1987	Fox	N/A	N/A
4826494	12/1988	Richmond et al.	N/A	N/A
4838883	12/1988	Matsuura	N/A	N/A
4840187	12/1988	Brazier	N/A	N/A
4863449	12/1988	Therriault et al.	N/A	N/A
4872450	12/1988	Austad	N/A	N/A
4878901	12/1988	Sachse	N/A	N/A
4897081	12/1989	Poirier et al.	N/A	N/A
4902565	12/1989	Brook	N/A	N/A
4906233	12/1989	Moriuchi et al.	N/A	N/A
4906240	12/1989	Reed et al.	N/A	N/A
4919654	12/1989	Kalt	N/A	N/A
4941882	12/1989	Ward et al.	N/A	N/A
4953565	12/1989	Tachibana et al.	N/A	N/A
4969880	12/1989	Zamierowski	N/A	N/A
4985019	12/1990	Michelson	N/A	N/A
5037397	12/1990	Kalt et al.	N/A	N/A
5086170	12/1991	Luheshi et al.	N/A	N/A
5092858	12/1991	Benson et al.	N/A	N/A
5100396	12/1991	Zamierowski	N/A	N/A
5134994	12/1991	Say	N/A	N/A
5149331	12/1991	Ferdman et al.	N/A	N/A
5167613	12/1991	Karami et al.	N/A	N/A
5176663	12/1992	Svedman et al.	N/A	N/A

5215522	12/1992	Page et al.	N/A	N/A
5232453	12/1992	Plass et al.	N/A	N/A
5261893	12/1992	Zamierowski	N/A	N/A
5278100	12/1993	Doan et al.	N/A	N/A
5279550	12/1993	Habib et al.	N/A	N/A
5298015	12/1993	Komatsuzaki et al.	N/A	N/A
5342376	12/1993	Ruff	N/A	N/A
5344415	12/1993	DeBusk et al.	N/A	N/A
5358494	12/1993	Svedman	N/A	N/A
5437622	12/1994	Carion	N/A	N/A
5437651	12/1994	Todd et al.	N/A	N/A
5527293	12/1995	Zamierowski	N/A	N/A
5549584	12/1995	Gross	N/A	N/A
5556375	12/1995	Ewall	N/A	N/A
5607388	12/1996	Ewall	N/A	N/A
5636643	12/1996	Argenta et al.	N/A	N/A
5645081	12/1996	Argenta et al.	N/A	N/A
6071267	12/1999	Zamierowski	N/A	N/A
6135116	12/1999	Vogel et al.	N/A	N/A
6241747	12/2000	Ruff	N/A	N/A
6287316	12/2000	Agarwal et al.	N/A	N/A
6345623	12/2001	Heaton et al.	N/A	N/A
6488643	12/2001	Tumey et al.	N/A	N/A
6493568	12/2001	Bell et al.	N/A	N/A
6553998	12/2002	Heaton et al.	N/A	N/A
6765123	12/2003	de Jong et al.	N/A	N/A
6814079	12/2003	Heaton et al.	N/A	N/A
7381859	12/2007	Hunt et al.	N/A	N/A
7846141	12/2009	Weston	N/A	N/A
7951124	12/2010	Boehringer et al.	N/A	N/A
8062273	12/2010	Weston	N/A	N/A
8216198	12/2011	Heagle et al.	N/A	N/A
8251979	12/2011	Malhi	N/A	N/A
8257327	12/2011	Blott et al.	N/A	N/A
8398614	12/2012	Blott et al.	N/A	N/A
8449509	12/2012	Weston	N/A	N/A
8529548	12/2012	Blott et al.	N/A	N/A
8535296	12/2012	Blott et al.	N/A	N/A
8551060	12/2012	Schuessler et al.	N/A	N/A
8568386	12/2012	Malhi	N/A	N/A
8679081	12/2013	Heagle et al.	N/A	N/A
8834451	12/2013	Blott et al.	N/A	N/A
8926592	12/2014	Blott et al.	N/A	N/A
9017302	12/2014	Vitaris et al.	N/A	N/A
9168180	12/2014	Ha et al.	N/A	N/A
9198801	12/2014	Weston	N/A	N/A
9211365	12/2014	Weston	N/A	N/A
9289542	12/2015	Blott et al.	N/A	N/A
9421309	12/2015	Robinson et al.	N/A	N/A

9918733	12/2017	Ingram et al.	N/A	N/A
9974694	12/2017	Locke	N/A	A61F 13/05
10369058	12/2018	Ha et al.	N/A	N/A
10610414	12/2019	Hartwell et al.	N/A	N/A
10736788	12/2019	Locke	N/A	A61F 13/01042
10743900	12/2019	Ingram et al.	N/A	N/A
11224542	12/2021	Robinson et al.	N/A	N/A
2001/0037118	12/2000	Shadduck	N/A	N/A
2002/0065494	12/2001	Lockwood et al.	N/A	N/A
2002/0077661	12/2001	Saadat	N/A	N/A
2002/0115951	12/2001	Norstrem et al.	N/A	N/A
2002/0120185	12/2001	Johnson	N/A	N/A
2002/0143286	12/2001	Tumey	N/A	N/A
2002/0161346	12/2001	Lockwood et al.	N/A	N/A
2004/0030304	12/2003	Hunt et al.	N/A	N/A
2005/0282895	12/2004	Dosch et al.	N/A	N/A
2007/0185426	12/2006	Ambrosio et al.	N/A	N/A
2008/0132819	12/2007	Radl et al.	N/A	N/A
2008/0177253	12/2007	Boehringer et al.	N/A	N/A
2008/0275409	12/2007	Kane et al.	N/A	N/A
2008/0300555	12/2007	Olson et al.	N/A	N/A
2009/0012482	12/2008	Pinto et al.	N/A	N/A
2009/0093779	12/2008	Riesinger	N/A	N/A
2009/0227969	12/2008	Jaeb	604/313	A61F 13/05
2009/0287129	12/2008	Boehringer	602/43	A61F 13/01042
2009/0299303	12/2008	Seegert	N/A	N/A
2010/0063484	12/2009	Heagle	N/A	N/A
2010/0160871	12/2009	Seegert et al.	N/A	N/A
2010/0160874	12/2009	Robinson et al.	N/A	N/A
2010/0160876	12/2009	Robinson et al.	N/A	N/A
2010/0185163	12/2009	Heagle	N/A	N/A
2010/0312159	12/2009	Aali et al.	N/A	N/A
2011/0054422	12/2010	Locke et al.	N/A	N/A
2011/0087176	12/2010	Blott et al.	N/A	N/A
2011/0118683	12/2010	Weston	N/A	N/A
2011/0178451	12/2010	Robinson et al.	N/A	N/A
2011/0213287	12/2010	Lattimore et al.	N/A	N/A
2011/0213319	12/2010	Blott et al.	N/A	N/A
2011/0230809	12/2010	Manwaring et al.	N/A	N/A
2011/0282309	12/2010	Adie	604/319	A61F 13/022
2011/0301556	12/2010	Lichtenstein	604/319	A61M 1/915
2012/0016334	12/2011	Nakajima et al.	N/A	N/A
2012/0143113	12/2011	Robinson et al.	N/A	N/A
2012/0143114	12/2011	Locke et al.	N/A	N/A
2012/0157945	12/2011	Robinson et al.	N/A	N/A
2013/0144230	12/2012	Wu	604/319	A61M 1/913

2013/0211349	12/2012	Stokes et al.	N/A	N/A
2013/0274717	12/2012	Dunn	N/A	N/A
2014/0066868	12/2013	Freedman et al.	N/A	N/A
2014/0155791	12/2013	Robinson et al.	N/A	N/A
2014/0163491	12/2013	Schuessler et al.	N/A	N/A
2014/0228787	12/2013	Croizat et al.	N/A	N/A
2014/0236112	12/2013	Von Wolff et al.	N/A	N/A
2015/0080788	12/2014	Blott et al.	N/A	N/A
2015/0119832	12/2014	Locke	N/A	N/A
2015/0174284	12/2014	Payne et al.	N/A	N/A
2015/0201954	12/2014	Pratt et al.	N/A	N/A
2015/0320434	12/2014	Ingram et al.	N/A	N/A
2015/0320602	12/2014	Locke	606/213	A61F 13/01017
2015/0320603	12/2014	Locke et al.	N/A	N/A
2016/0022885	12/2015	Dunn et al.	N/A	N/A
2016/0038345	12/2015	Ha et al.	N/A	N/A
2016/0095754	12/2015	Andrews et al.	N/A	N/A
2016/0158066	12/2015	Chao	N/A	N/A
2016/0354086	12/2015	Dunn	N/A	N/A
2017/0007462	12/2016	Hartwell et al.	N/A	N/A
2017/0135862	12/2016	Tuck et al.	N/A	N/A
2017/0189237	12/2016	Locke et al.	N/A	N/A
2017/0197006	12/2016	Johnson et al.	N/A	N/A
2017/0231822	12/2016	Hoggarth et al.	N/A	N/A
2017/0239095	12/2016	Hoggarth et al.	N/A	N/A
2018/0235646	12/2017	Locke et al.	N/A	N/A
2018/0353337	12/2017	Locke	N/A	N/A
2018/0353338	12/2017	Locke et al.	N/A	N/A
2018/0353662	12/2017	Locke et al.	N/A	N/A
2019/0117465	12/2018	Osborne et al.	N/A	N/A
2019/0192350	12/2018	Gowans et al.	N/A	N/A
2019/0231944	12/2018	Dunn et al.	N/A	N/A
2019/0274889	12/2018	Steward et al.	N/A	N/A
2019/0314209	12/2018	Ha et al.	N/A	N/A
2020/0046567	12/2019	Carroll et al.	N/A	N/A
2020/0107965	12/2019	Greener	N/A	N/A
2020/0146896	12/2019	Rice et al.	N/A	N/A
2020/0155359	12/2019	Carroll et al.	N/A	N/A
2020/0383837	12/2019	Gowans et al.	N/A	N/A
2021/0077302	12/2020	Carroll et al.	N/A	N/A
2021/0228417	12/2020	Ha et al.	N/A	N/A
2023/0000687	12/2022	Rice et al.	N/A	N/A
2023/0000688	12/2022	Rice et al.	N/A	N/A
2024/0099898	12/2023	Rice et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	**Application Date**	**Country**	**CPC**
550575	12/1985	AU	N/A

745271	12/2001	AU	N/A
755496	12/2001	AU	N/A
2005436	12/1989	CA	N/A
26 40 413	12/1977	DE	N/A
43 06 478	12/1993	DE	N/A
29 504 378	12/1994	DE	N/A
102006017194	12/2006	DE	N/A
0100148	12/1983	EP	N/A
0117632	12/1983	EP	N/A
0161865	12/1984	EP	N/A
0358302	12/1989	EP	N/A
1018967	12/1999	EP	N/A
2098257	12/2008	EP	N/A
3263079	12/2017	EP	N/A
3378450	12/2017	EP	N/A
692578	12/1952	GB	N/A
2195255	12/1987	GB	N/A
2 197 789	12/1987	GB	N/A
2 220 357	12/1989	GB	N/A
2 235 877	12/1990	GB	N/A
2 329 127	12/1998	GB	N/A
2 333 965	12/1998	GB	N/A
2365350	12/2001	GB	N/A
2377939	12/2002	GB	N/A
S57-013040	12/1981	JP	N/A
4129536	12/2007	JP	N/A
2008136998	12/2007	NO	N/A
71559	12/2001	SG	N/A
80/02182	12/1979	WO	N/A
87/04626	12/1986	WO	N/A
90/010424	12/1989	WO	N/A
93/009727	12/1992	WO	N/A
94/20041	12/1993	WO	N/A
96/05873	12/1995	WO	N/A
97/18007	12/1996	WO	N/A
99/13793	12/1998	WO	N/A
0185248	12/2000	WO	N/A
2005102234	12/2004	WO	N/A
2006114638	12/2005	WO	N/A
2008/005532	12/2007	WO	N/A
2009021523	12/2008	WO	N/A
2009/146441	12/2008	WO	N/A
2010051071	12/2009	WO	N/A
2010051073	12/2009	WO	N/A
2010075178	12/2009	WO	N/A
2011008497	12/2010	WO	N/A
2011089098	12/2010	WO	N/A
2013/032745	12/2012	WO	N/A
2013066426	12/2012	WO	N/A
2013071243	12/2012	WO	N/A

2013116552	12/2012	WO	N/A
2013129343	12/2012	WO	N/A
2013149078	12/2012	WO	N/A
2014014871	12/2013	WO	N/A
WO-2014014922	12/2013	WO	A61F 13/00068
2014024048	12/2013	WO	N/A
2014143487	12/2013	WO	N/A
2015172104	12/2014	WO	N/A
2015172111	12/2014	WO	N/A
2015173547	12/2014	WO	N/A
2017195038	12/2016	WO	N/A
2018/077872	12/2017	WO	N/A
2018/094061	12/2017	WO	N/A
2018/226328	12/2017	WO	N/A
2019136164	12/2018	WO	N/A
2019152422	12/2018	WO	N/A
2020097529	12/2019	WO	N/A

OTHER PUBLICATIONS

International Search Report and Written Opinion for PCT/US2015/030027 mailed Jul. 15, 2015. cited by applicant
Partial International Search Report from PCT/US2015/030030 mailed Jul. 22, 2015. cited by applicant
International Search Report and Written Opinion for PCT/US2015/030023 mailed Aug. 24, 2015. cited by applicant
Extended European Search Report for corresponding Application No. 171862527, mailed Nov. 14, 2017. cited by applicant
“Introduction to Polyurethanes: Thermoplastic Polyurethane”, American Chemistry Council, 2018, https://polyurethane.americanchemistry.com/polyurethanes/Introduction-to-Polyurethanes/Applications/Thermoplastic-Polyurethane/. cited by applicant
International Search Report and Written Opinion for corresponding Application No. PCT/US2019/027463, mailed Jul. 4, 2019. cited by applicant
Japanese Notice of Rejection for corresponding Application No. 2016-566815, mailed Feb. 5, 2019. cited by applicant
Extended European Search Report for corresponding Application No. 18162504.7, mailed May 24, 2018. cited by applicant
Japanese Notice of Rejection for corresponding Application No. 2016-566785, mailed Jun. 25, 2019. cited by applicant
Japanese Notice of Rejection for corresponding Application No. 2016-566785, mailed Jan. 29, 2019. cited by applicant
Non-Final Office Action for Corresponding U.S. Appl. No. 15/960,310, mailed Apr. 29, 2020. cited by applicant
Japanese Notice of Rejection for Corresponding Application No. 2019-233695, mailed Oct. 13, 2020. cited by applicant
International Search Report and Written Opinion for Corresponding Application No. PCT/US2019/060567, mailed Feb. 14, 2020. cited by applicant
International Search Report and Written Opinion for Corresponding Application No. PCT/US2019/045505, mailed Nov. 7, 2019. cited by applicant
International Search Report and Written Opinion for Corresponding Application No.

PCT/US2019/060479, mailed Apr. 7, 2020. cited by applicant
International Search Report and Written Opinion for Corresponding Application No. PCT/US2020/013922, mailed May 4, 2020. cited by applicant
Chinese Notice of Rejection Corresponding to Application No. 2020800099951, mailed Mar. 28, 2022. cited by applicant
International Search Report and Written Opinion for Corresponding Application No. PCT/IB2020/056911, mailed Oct. 21, 2020. cited by applicant
International Search Report and Written Opinion Corresponding to Application No. PCT/IB2020/061435, mailed Mar. 16, 2021. cited by applicant
International Search Report and Written Opinion Corresponding to Application No. PCT/IB2020/061465, mailed Mar. 16, 2021. cited by applicant
International Search Report and Written Opinion Corresponding to Application No. PCT/IB2020/061540, mailed Feb. 24, 2021. cited by applicant
Canadian Examination Report for related application 2,947,302, dated Jun. 11, 2021. cited by applicant
Japanese Notice of Rejection for related application 2020-557257, dated Feb. 28, 2023. cited by applicant
Louis C. Argenta, MD and Michael J. Morykwas, PHD; Vacuum-Assisted Closure: A New Method for Wound Control and Treatment: Clinical Experience; Annals of Plastic Surgery; vol. 38, No. 6, Jun. 1997; pp. 563-576. cited by applicant
Susan Mendez-Eatmen, RN; “When wounds Won't Heal” RN Jan. 1998, vol. 61 (1); Medical Economics Company, Inc., Montvale, NJ, USA; pp. 20-24. cited by applicant
James H. Blackburn II, MD et al.: Negative-Pressure Dressings as a Bolster for Skin Grafts; Annals of Plastic Surgery, vol. 40, No. 5, May 1998, pp. 453-457; Lippincott Williams & Wilkins, Inc., Philidelphia, PA, USA. cited by applicant
John Masters; “Reliable, Inexpensive and Simple Suction Dressings”; Letter to the Editor, British Journal of Plastic Surgery, 1998, vol. 51 (3), p. 267; Elsevier Science/The British Association of Plastic Surgeons, UK. cited by applicant
S.E. Greer, et al. “The Use of Subatmospheric Pressure Dressing Therapy to Close Lymphocutaneous Fistulas of the Groin” British Journal of Plastic Surgery (2000), 53, pp. 484-487. cited by applicant
George V. Letsou, MD., et al.; “Stimulation of Adenylate Cyclase Activity in Cultured Endothelial Cells Subjected to Cyclic Stretch”; Journal of Cardiovascular Surgery, 31, 1990, pp. 634-639. cited by applicant
Orringer, Jay, et al.; “Management of Wounds in Patients with Complex Enterocutaneous Fistulas”; Surgery, Gynecology & Obstetrics, Jul. 1987, vol. 165, pp. 79-80. cited by applicant
International Search Report for PCT International Application PCT/GB95/01983; Nov. 23, 1995. cited by applicant
PCT International Search Report for PCT International Application PCT/GB98/02713; Jan. 8, 1999. cited by applicant
PCT Written Opinion; PCT International Application PCT/GB98/02713; Jun. 8, 1999. cited by applicant
PCT International Examination and Search Report, PCT International Application PCT/GB96/02802; Jan. 15, 1998 & Apr. 29, 1997. cited by applicant
PCT Written Opinion, PCT International Application PCT/GB96/02802; Sep. 3, 1997. cited by applicant
Dattilo, Philip P., Jr., et al.; “Medical Textiles: Application of an Absorbable Barbed Bi-directional Surgical Suture”; Journal of Textile and Apparel, Technology and Management, vol. 2, Issue 2, Spring 2002, pp. 1-5. cited by applicant
Kostyuchenok, B.M., et al.; “Vacuum Treatment in the Surgical Management of Purulent Wounds”;

Vestnik Khirurgi, Sep. 1986, pp. 18-21 and 6 page English translation thereof. cited by applicant
Davydov, Yu. A., et al.; “Vacuum Therapy in the Treatment of Purulent Lactation Mastitis”; Vestnik Khirurgi, May 14, 1986, pp. 66-70, and 9 page English translation thereof. cited by applicant
Yusupov. Yu.N., et al.; “Active Wound Drainage”, Vestnki Khirurgi, vol. 138, Issue 4, 1987, and 7 page English translation thereof. cited by applicant
Davydov, Yu.A., et al.; “Bacteriological and Cytological Assessment of Vacuum Therapy for Purulent Wounds”; Vestnik Khirugi, Oct. 1988, pp. 48-52, and 8 page English translation thereof. cited by applicant
Davydov, Yu.A., et al.; “Concepts for the Clinical-Biological Management of the Wound Process in the Treatment of Purulent Wounds by Means of Vacuum Therapy”; Vestnik Khirurgi, Jul. 7, 1980, pp. 132-136, and 8 page English translation thereof. cited by applicant
Chariker, Mark E., M.D., et al.; “Effective Management of incisional and cutaneous fistulae with closed suction wound drainage”; Contemporary Surgery, vol. 34, Jun. 1989, pp. 59-63. cited by applicant
Egnell Minor, Instruction Book, First Edition, 300 7502, Feb. 1975, pp. 24. cited by applicant
Egnell Minor: Addition to the Users Manual Concerning Overflow Protection—Concerns all Egnell Pumps, Feb. 3, 1983, pp. 2. cited by applicant
Svedman, P. “Irrigation Treatment of Leg Ulcers”, The Lancet, Sep. 3, 1983, pp. 532-534. cited by applicant
Chinn, Steven D. et al.: “Closed Wound Suction Drainage”, The Journal of Foot Surgery, vol. 24, No. 1, 1985, pp. 76-81. cited by applicant
Arnljots, Bjöm et al.: “Irrigation Treatment in Split-Thickness Skin Grafting of Intractable Leg Ulcers”, Scand J. Plast Reconstr. Surg., No. 19, 1985, pp. 211-213. cited by applicant
Svedman, P.: “A Dressing Allowing Continuous Treatment of a Biosurface”, IRCS Medical Science: Biomedical Technology, Clinical Medicine, Surgery and Transplantation, vol. 7, 1979, p. 221. cited by applicant
Svedman, P. et al: “A Dressing System Providing Fluid Supply and Suction Drainage Used for Continuous of Intermittent Irrigation”, Annals of Plastic Surgery, vol. 17, No. 2, Aug. 1986, pp. 125-133. cited by applicant
N.A. Bagautdinov, “Variant of External Vacuum Aspiration in the Treatment of Purulent Diseases of Soft Tissues,” Current Problems in Modern Clinical Surgery: Interdepartmental Collection, edited by V. Ye Volkov et al. (Chuvashia State University, Cheboksary, U.S.S.R. 1986); pp. 94-96 (copy and certified translation). cited by applicant
K.F. Jeter, T.E. Tintle, and M. Chariker, “Managing Draining Wounds and Fistulae: New and Established Methods,” Chronic Wound Care, edited by D. Krasner (Health Management Publications, Inc., King of Prussia, PA 1990), pp. 240-246. cited by applicant
G. Živadinovi?, V. ?uki?, Ž. Maksimovi?, ?. Radak, and P. Peška, “Vacuum Therapy in the Treatment of Peripheral Blood Vessels,” Timok Medical Journal 11 (1986), pp. 161-164 (copy and certified translation). cited by applicant
F.E. Johnson, “An Improved Technique for Skin Graft Placement Using a Suction Drain,” Surgery, Gynecology, and Obstetrics 159 (1984), pp. 584-585. cited by applicant
A.A. Safronov, Dissertation Abstract, Vacuum Therapy of Trophic Ulcers of the Lower Leg with Simultaneous Autoplasty of the Skin (Central Scientific Research Institute of Traumatology and Orthopedics, Moscow, U.S.S.R. 1967) (copy and certified translation). cited by applicant
M. Schein, R. Saadia, J.R. Jamieson, and G.A.G. Decker, “The ‘Sandwich Technique’ in the Management of the Open Abdomen,” British Journal of Surgery 73 (1986), pp. 369-370. cited by applicant
D.E. Tribble, An Improved Sump Drain-Irrigation Device of Simple Construction, Archives of Surgery 105 (1972) pp. 511-513. cited by applicant
M.J. Morykwas, L.C. Argenta, E.I. Shelton-Brown, and W. McGuirt, “Vacuum-Assisted Closure: A

New Method for Wound Control and Treatment: Animal Studies and Basic Foundation," Annals of Plastic Surgery 38 (1997), pp. 553-562 (Morykwas I). cited by applicant
C.E. Tennants, "The Use of Hypermia in the Postoperative Treatment of Lesions of the Extremities and Thorax," Journal of the American Medical Association 64 (1915), pp. 1548-1549. cited by applicant
Selections from W. Meyer and V. Schmieden, Bier's Hyperemic Treatment in Surgery, Medicine, and the Specialties: A Manual of Its Practical Application, (W.B. Saunders Co., Philadelphia, PA 1909), pp. 17-25, 44-64, 90-96, 167-170, and 210-211. cited by applicant
V.A. Solovev et al., Guidelines, The Method of Treatment of Immature External Fistulas in the Upper Gastrointestinal Tract, editor-in-chief Prov. V.I. Parahonyak (S.M. Kirov Gorky State Medical Institute, Gorky, U.S.S.R. 1987) ("Solovev Guidelines"). cited by applicant
V.A. Kuznetsov & N.a. Bagautdinov, "Vacuum and Vacuum-Sorption Treatment of Open Septic Wounds," in II All-Union Conference on Wounds and Wound Infections: Presentation Abstracts, edited by B.M. Kostyuchenok et al. (Moscow, U.S.S.R. Oct. 28-29, 1986) pp. 91-92 ("Bagautdinov II"). cited by applicant
V.A. Solovev, Dissertation Abstract, Treatment and Prevention of Suture Failures after Gastric Resection (S.M. Kirov Gorky State Medical Institute, Gorky, U.S.S.R. 1988) ("Solovev Abstract"). cited by applicant
V.A.C. ® Therapy Clinical Guidelines: A Reference Source for Clinicians; Jul. 2007. cited by applicant
Office Action for related U.S. Appl. No. 16/678,450, dated Jul. 31, 2023, 22 pages. cited by applicant
Office Action for related U.S. Appl. No. 16/923,651, dated Aug. 28, 2023, 27 pages. cited by applicant
Japanese Notice of Rejection for Application No. 2021-542412, dated Dec. 5, 2023. cited by applicant
European Examination Report for Application No. 20747133.5, dated Dec. 21, 2023. cited by applicant
Office action for related U.S. Appl. No. 16/923,651 dated Feb. 12, 2024. cited by applicant
Japanese Notice of Rejection for Application No. 2019-233695 dated Mar. 5, 2024. cited by applicant
Office action for related Application No. 17/629, 174, dated Mar. 26, 2024. cited by applicant
Japanese Notice of Rejection for Application No. 2021-524440 dated Apr. 16, 2024. cited by applicant
Office action for U.S. Appl. No. 16/678,450, dated Sep. 9, 2024. cited by applicant
Office action for U.S. Appl. No. 16/745,075, dated Jul. 24, 2024. cited by applicant
Japanese Decision of Rejection and Decision for Dismissal of Amendment for Application No. 2021-524440, dated Oct. 15, 2024. cited by applicant
Copper Development Association Inc., Introduction to Antimicrobial Copper, Feb. 15, 2024. cited by applicant
Office action for U.S. Appl. No. 17/629,174, dated Feb. 26, 2025. cited by applicant
Office action for U.S. Appl. No. 17/779,755, dated Apr. 9, 2025. cited by applicant
Office action for U.S. Appl. No. 17/779,792, dated Jun. 3, 2025. cited by applicant

---

Primary Examiner: Tanner; Jocelin C

Background/Summary

RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 15/958,773, filed Apr. 20, 2018, which is a continuation of U.S. patent application Ser. No. 14/708,061, filed May 8, 2015, now U.S. Pat. No. 9,974,694, which claims the benefit, under 35 USC § 119(e), of U.S. Provisional Patent Application No. 61/991,174, entitled "Dressing with Contracting Layer for Linear Tissue Sites," filed May 9, 2014, which are incorporated herein by reference for all purposes.

TECHNICAL FIELD

(1) The invention set forth in the appended claims relates generally to tissue treatment systems and more particularly, but without limitation, to a dressing having a contracting layer for assisting in closure of linear tissue sites.

BACKGROUND

(2) Clinical studies and practice have shown that reducing pressure in proximity to a tissue site can augment and accelerate growth of new tissue at the tissue site. The applications of this phenomenon are numerous, but it has proven particularly advantageous for treating wounds. Regardless of the etiology of a wound, whether trauma, surgery, or another cause, proper care of the wound is important to the outcome. Treatment of wounds or other tissue with reduced pressure may be commonly referred to as "negative-pressure therapy," but is also known by other names, including "negative-pressure wound therapy," "reduced-pressure therapy," "vacuum therapy," and "vacuum-assisted closure," for example. Negative-pressure therapy may provide a number of benefits, including migration of epithelial and subcutaneous tissues, improved blood flow, and micro-deformation of tissue at a wound site. Together, these benefits can increase development of granulation tissue and reduce healing times.

(3) While the clinical benefits of negative-pressure therapy are widely known, the cost and complexity of negative-pressure therapy can be a limiting factor in its application, and the development and operation of negative-pressure systems, components, and processes continues to present significant challenges to manufacturers, healthcare providers, and patients.

BRIEF SUMMARY

(4) New and useful systems, apparatuses, and methods for closing an opening through a surface of a tissue site are set forth in the appended claims. Illustrative embodiments are also provided to enable a person skilled in the art to make and use the claimed subject matter. For example, a system for closing an opening through a surface of a tissue site is described. The system may include a sealing member adapted to cover the opening to form a sealed space and a negative-pressure source adapted to be fluidly coupled to the sealed space to provide negative pressure to the sealed space. The system can include a protective layer adapted to be positioned adjacent to the opening. The system may also include a contracting layer adapted to be positioned adjacent to the protective layer and formed from a material having a firmness factor and a plurality of holes extending through the contracting layer to form a void space. The holes may have a perforation shape factor and a strut angle causing the plurality of holes to collapse in a direction substantially perpendicular to the opening. The contracting layer may generate a closing force substantially parallel to the surface of the tissue site to close the opening in response to application of the negative pressure.

(5) Alternatively, other example embodiments may include an apparatus for closing an opening through a surface of a tissue site. The apparatus may include a contracting layer adapted to be positioned adjacent the opening and formed from a material having a firmness factor and a plurality of holes extending through the contracting layer to form a void space. The holes may have a perforation shape factor and a strut angle causing the plurality of holes to collapse in a direction substantially perpendicular to the opening. The contracting layer may generate a closing force substantially parallel to the surface of the tissue site to close the opening in response to application of a negative pressure.

(6) A method for closing an opening through a surface of a tissue site is also described. The method

may include positioning a contracting layer adjacent to and covering the tissue site. The contracting layer may be adapted to be positioned adjacent the opening and formed from a material having a firmness factor and a plurality of holes extending through the contracting layer to form a void space. The holes may have a perforation shape factor and a strut angle causing the plurality of holes to collapse in a direction substantially perpendicular to the opening. The contracting layer may be collapsed parallel to the surface of the tissue site to generate a closing force.
(7) Objectives, advantages, and a preferred mode of making and using the claimed subject matter may be understood best by reference to the accompanying drawings in conjunction with the following detailed description of illustrative embodiments.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a perspective section view with a portion show in elevation, illustrating details that may be associated with some embodiments of a negative-pressure therapy system;
(2) FIG. **2** is a plan view, illustrating details that may be associated with some embodiments of a contracting layer of the negative-pressure therapy system of FIG. **1** in a first position;
(3) FIG. **3** is a schematic view, illustrating details that may be associated with some embodiments of a hole of the contracting layer of FIG. **2**;
(4) FIG. **4** is a plan view, illustrating details that may be associated with some embodiments of the holes of the contracting layer of FIG. **2**;
(5) FIG. **5** is a plan view, illustrating details that may be associated with some embodiments of the contracting layer of FIG. **2** in a second position;
(6) FIG. **6** is a perspective section view with a portion show in elevation, illustrating details that may be associated with some embodiments of the negative-pressure therapy system of FIG. **1**;
(7) FIG. **7** is a plan view, illustrating details that may be associated with some embodiments of another contracting layer of the negative-pressure therapy system of FIG. **1**;
(8) FIG. **8**A is a schematic view, illustrating details that may be associated with some embodiments of a hole of the contracting layer of FIG. **7**;
(9) FIG. **8**B is a plan view, illustrating details that may be associated with some embodiments of the holes of the contracting layer of FIG. **7**;
(10) FIG. **9**A is a plan view, illustrating details that may be associated with some embodiments of another contracting layer of the negative-pressure therapy system of FIG. **1**;
(11) FIG. **9**B is a plan view, illustrating details that may be associated with some embodiments of the holes of the contracting layer of FIG. **9**A;
(12) FIG. **10** is a schematic view, illustrating details that may be associated with some embodiments of a hole of the contracting layer of FIG. **9**A having a perforation shape factor;
(13) FIG. **11** is a schematic view, illustrating details that may be associated with some embodiments of a hole of the contracting layer of FIG. **9**A having another perforation shape factor;
(14) FIG. **12** is a schematic view, illustrating details that may be associated with some embodiments of a hole of the contracting layer of FIG. **9**A having another perforation shape factor;
(15) FIG. **13**A is a plan view, illustrating details that may be associated with some embodiments of another contracting layer of the negative-pressure therapy system of FIG. **1**;
(16) FIG. **13**B is a plan view, illustrating details that may be associated with some embodiments of the holes of the contracting layer of FIG. **13**A;
(17) FIG. **14** is a schematic view, illustrating details that may be associated with some embodiments of a hole of the contracting layer of FIG. **13**A; and
(18) FIG. **15** is a plan view, illustrating details that may be associated with some embodiments of another contracting layer of the negative-pressure therapy system of FIG. **1**.

DESCRIPTION OF EXAMPLE EMBODIMENTS

(19) The following description of example embodiments provides information that enables a person skilled in the art to make and use the subject matter set forth in the appended claims, but may omit certain details already well-known in the art. The following detailed description is, therefore, to be taken as illustrative and not limiting.

(20) The example embodiments may also be described herein with reference to spatial relationships between various elements or to the spatial orientation of various elements depicted in the attached drawings. In general, such relationships or orientation assume a frame of reference consistent with or relative to a patient in a position to receive treatment. However, as should be recognized by those skilled in the art, this frame of reference is merely a descriptive expedient rather than a strict prescription.

(21) FIG. **1** illustrates details that may be associated with some embodiments of a negative-pressure therapy system **100** that can be applied to a tissue site **102**. In some embodiments, the tissue site **102** may be closed by one or more stitches **103**. The negative-pressure therapy system **100** may include a dressing and a negative-pressure source. For example, a dressing **104** may be fluidly coupled to a negative-pressure source **106**, as illustrated in FIG. **1**. A dressing may be fluidly coupled to a negative-pressure source by a connector and a tube. The dressing **104**, for example, may be fluidly coupled to the negative-pressure source **106** by a connector **108** and a tube **110**. A dressing may generally include a cover and a tissue interface. The dressing **104**, for example, may include a cover **112** and a tissue interface **113**. In some embodiments, the tissue interface **113** may include a contracting layer, such as a contracting layer **114**, and a protective layer, such as a protective layer **116**.

(22) In general, components of the negative-pressure therapy system **100** may be coupled directly or indirectly. For example, the negative-pressure source **106** may be directly coupled to the dressing **104** and indirectly coupled to the tissue site **102** through the dressing **104**. Components may be fluidly coupled to each other to provide a path for transferring fluids (i.e., liquid and/or gas) between the components.

(23) In some embodiments, components may be fluidly coupled through a tube, such as the tube **110**. A “tube,” as used herein, broadly refers to a tube, pipe, hose, conduit, or other structure with one or more lumina adapted to convey a fluid between two ends. Typically, a tube is an elongated, cylindrical structure with some flexibility, but the geometry and rigidity may vary. In some embodiments, components may additionally or alternatively be coupled by virtue of physical proximity, being integral to a single structure, or being formed from the same piece of material. Coupling may also include mechanical, thermal, electrical, or chemical coupling (such as a chemical bond) in some contexts.

(24) In operation, a tissue interface, such as the tissue interface **113**, may be placed within, over, on, or otherwise proximate to a tissue site. A cover may be placed over a tissue interface and sealed to tissue near a tissue site. For example, the tissue interface **113** may be placed over the tissue site **102**, and the cover **112** may be sealed to undamaged epidermis peripheral to the tissue site **102**. Thus, the cover **112** can provide a sealed therapeutic environment **118** proximate to the tissue site **102** that is substantially isolated from the external environment, and the negative-pressure source **106** can reduce the pressure in the sealed therapeutic environment **118**.

(25) The fluid mechanics of using a negative-pressure source to reduce pressure in another component or location, such as within a sealed therapeutic environment, can be mathematically complex. However, the basic principles of fluid mechanics applicable to negative-pressure therapy are generally well-known to those skilled in the art, and the process of reducing pressure may be described illustratively herein as “delivering,” “distributing,” or “generating” negative pressure, for example.

(26) In general, exudates and other fluids flow toward lower pressure along a fluid path. Thus, the term “downstream” typically refers to a position in a fluid path relatively closer to a negative-

pressure source. Conversely, the term “upstream” refers to a position relatively further away from a negative-pressure source. Similarly, it may be convenient to describe certain features in terms of fluid “inlet” or “outlet” in such a frame of reference. This orientation is generally presumed for purposes of describing various features and components of negative-pressure therapy systems herein. However, the fluid path may also be reversed in some applications (such as by substituting a positive-pressure source for a negative-pressure source) and this descriptive convention should not be construed as a limiting convention.

(27) The term “tissue site” in this context broadly refers to a wound or a defect located on or within tissue, including but not limited to, bone tissue, adipose tissue, muscle tissue, neural tissue, dermal tissue, vascular tissue, connective tissue, cartilage, tendons, or ligaments. A wound may include chronic, acute, traumatic, subacute, and dehisced wounds, partial-thickness burns, ulcers (such as diabetic, pressure, or venous insufficiency ulcers), flaps, and grafts, for example. The term “tissue site” may also refer to areas of any tissue that are not necessarily wounded or defective, but are instead areas in which it may be desirable to add or promote the growth of additional tissue. For example, negative pressure may be used at a tissue site to grow additional tissue that may be harvested and transplanted to a tissue site at another location.

(28) A tissue site may also be characterized by shape. For example, some tissue sites may be referred to as a linear tissue site. A linear tissue site may generally refer to a tissue site having an elongated shape, such as an incision having a length substantially greater than its width. An incision may have edges that may be substantially parallel, particularly if the incision is caused by a scalpel, knife, razor, or other sharp blade. Other examples of a linear tissue site may include a laceration, a puncture, or other separation of tissue, which may have been caused by trauma, surgery, or degeneration. In some embodiments, a linear tissue site may also be an incision in an organ adjacent a fistula. In some embodiments, a linear tissue site may be an incision or puncture in otherwise healthy tissue that extends up to 40 cm or more in length. In some embodiments, a linear tissue site may also vary in depth. For example, an incision may have a depth that extends up to 15 cm or more or may be subcutaneous depending on the type of tissue and the cause of the incision.

(29) The tissue site **102**, for example, may illustrate a linear tissue site having a tissue surface **105** and an opening **120** through the tissue surface **105** along a length of the tissue site **102**. The tissue site **102** may also have a first wall **122** and a second wall **124** extending from the opening **120** in the tissue surface **105** generally parallel to each other along the length and depth of the tissue site **102**.

(30) In some embodiments, one or more stitches **103** may be used to close the opening **120**. The stitches **103** may be surgical sutures, for example, which may be used to hold tissue together following an injury or a surgical procedure. Generally, stitches may be thread formed from absorbable material such as polyglycolic acid, polylactic acid, monocryls, and polydioxanone, or non-absorbable materials such as nylon, polyester, polyvinylidene fluoride, and polypropylene. The stitches **103** may apply a closing force to the opening **120** by being placed under tension to draw the first wall **122** and the second wall **124** toward each other.

(31) “Negative pressure” generally refers to a pressure less than a local ambient pressure. A local ambient pressure may be a pressure in a local environment external to the sealed therapeutic environment **118** provided by the dressing **104**. In many cases, a local ambient pressure may also be the atmospheric pressure at which a tissue site is located. Alternatively, negative pressure may be a pressure that is less than a hydrostatic pressure associated with tissue at the tissue site. Unless otherwise indicated, values of pressure stated herein are gauge pressures. Similarly, references to increases in negative pressure typically refer to a decrease in absolute pressure, while decreases in negative pressure typically refer to an increase in absolute pressure.

(32) A negative-pressure source, such as the negative-pressure source **106**, may be a reservoir of air at a negative pressure, or may be a manual or electrically-powered device that can reduce the pressure in a sealed volume, such as a vacuum pump, a suction pump, a wall suction port available

at many healthcare facilities, or a micro-pump, for example. A negative-pressure source may be housed within or used in conjunction with other components, such as sensors, processing units, alarm indicators, memory, databases, software, display devices, or user interfaces that further facilitate negative-pressure therapy. While the amount and nature of negative pressure applied to a tissue site may vary according to therapeutic requirements, the pressure is generally a low vacuum, also referred to as a rough vacuum, between −5 mm Hg (−667 Pa) and −500 mm Hg (−66.7 kPa). Common therapeutic ranges are between −75 mm Hg (−9.9 kPa) and −300 mm Hg (−39.9 kPa).

(33) A tissue interface, such as the contracting layer **114** or the protective layer **116**, can generally be adapted to contact a tissue site. A tissue interface may be partially or fully in contact with a tissue site. If a tissue site is a wound, for example, a tissue interface may partially or completely fill the wound, or may be placed over the wound. A tissue interface may take many forms, and may have many sizes, shapes, or thicknesses depending on a variety of factors, such as the type of treatment being implemented or the nature and size of a tissue site. For example, the size and shape of a tissue interface may be adapted to the contours of deep and irregular shaped tissue sites.

(34) In some embodiments, a tissue interface may be a manifold or may include a manifold. A “manifold” in this context generally includes any substance or structure providing a plurality of pathways adapted to collect or distribute fluid across a tissue site under negative pressure. For example, a manifold may be adapted to receive negative pressure from a source and distribute the negative pressure through multiple apertures across a tissue site, which may have the effect of collecting fluid from across a tissue site and drawing the fluid toward the source. In some embodiments, the fluid path may be reversed or a secondary fluid path may be provided to facilitate delivering fluid across a tissue site.

(35) In some illustrative embodiments, the pathways of a manifold may be channels interconnected to improve distribution or collection of fluids across a tissue site. For example, cellular foam, open-cell foam, reticulated foam, porous tissue collections, and other porous material such as gauze or felted mat generally include pores, edges, and/or walls adapted to form interconnected fluid pathways. Liquids, gels, and other foams may also include or be cured to include apertures and flow channels. In some illustrative embodiments, a manifold may be a porous foam material having interconnected cells or pores adapted to uniformly (or quasi-uniformly) distribute negative pressure to a tissue site. The foam material may be either hydrophobic or hydrophilic. In one non-limiting example, a manifold may be an open-cell, reticulated polyurethane foam such as GranuFoam® dressing available from Kinetic Concepts, Inc. of San Antonio, Texas.

(36) In an example in which a tissue interface may be made from a hydrophilic material, the tissue interface may also wick fluid away from a tissue site, while continuing to distribute negative pressure to the tissue site. The wicking properties of a tissue interface may draw fluid away from a tissue site by capillary flow or other wicking mechanisms. An example of a hydrophilic foam is a polyvinyl alcohol, open-cell foam such as V.A.C. WhiteFoam® dressing available from Kinetic Concepts, Inc. of San Antonio, Texas Other hydrophilic foams may include those made from polyether. Other foams that may exhibit hydrophilic characteristics include hydrophobic foams that have been treated or coated to provide hydrophilicity.

(37) A tissue interface may further promote granulation at a tissue site when pressure within the sealed therapeutic environment is reduced. For example, any or all of the surfaces of a tissue interface may have an uneven, coarse, or jagged profile that can induce microstrains and stresses at a tissue site if negative pressure is applied through a tissue interface. In some embodiments, a tissue interface may be constructed from bioresorbable materials. Suitable bioresorbable materials may include, without limitation, a polymeric blend of polylactic acid (PLA) and polyglycolic acid (PGA). The polymeric blend may also include without limitation polycarbonates, polyfumarates, and capralactones. A tissue interface may further serve as a scaffold for new cell-growth, or a scaffold material may be used in conjunction with the tissue interface to promote cell-growth. A scaffold is generally a substance or structure used to enhance or promote the growth of cells or

formation of tissue, such as a three-dimensional porous structure that provides a template for cell growth. Illustrative examples of scaffold materials include calcium phosphate, collagen, PLA/PGA, coral hydroxy apatites, carbonates, or processed allograft materials.
(38) In some embodiments, the cover **112** may provide a bacterial barrier and protection from physical trauma. The cover **112** may also be constructed from a material that can reduce evaporative losses and provide a fluid seal between two components or two environments, such as between a therapeutic environment and a local external environment. The cover **112** may be, for example, an elastomeric film or membrane that can provide a seal adequate to maintain a negative pressure at a tissue site for a given negative-pressure source. In some example embodiments, the cover **112** may be a polymer drape, such as a polyurethane film, that is permeable to water vapor but impermeable to liquid. Such drapes typically have a thickness in the range of about 25 microns to about 50 microns. For permeable materials, the permeability generally should be low enough that a desired negative pressure may be maintained.
(39) An attachment device may be used to attach the cover **112** to an attachment surface, such as undamaged epidermis, a gasket, or another cover. In some embodiments, an attachment surface may be tissue surrounding a tissue site, such as the tissue surface **105** surrounding the opening **120**. An attachment device may take many forms. For example, an attachment device may be a medically-acceptable, pressure-sensitive adhesive that extends about a periphery, a portion, or an entire sealing member. In some embodiments, for example, some or all of the cover **112** may be coated with an acrylic adhesive having a coating weight between about 25 grams per square meter (gsm) and about 65 gsm. Thicker adhesives or combinations of adhesives may be applied in some embodiments to improve the seal and reduce leaks. Other example embodiments of an attachment device may include a double-sided tape, paste, hydrocolloid, hydrogel, silicone gel, or organogel.
(40) A linear tissue site, such as an incision, may often be created during a surgical procedure if a surgeon or other clinician uses a cutting instrument, such as a scalpel, to pierce and cut through at least a portion of a tissue site. Following a surgical procedure, a closing force may be applied to an opening of an incision to facilitate healing. A closing force may be a force that is substantially parallel to the tissue surface **105** and urges the first wall **122** and the second wall **124** toward each other to close the opening **120**. Closure of an opening may help maintain a healing environment for internal structures of a tissue site, as well as inhibit entry of bacteria or other harmful substances into the tissue site.
(41) Often, a mechanical means may be used to apply a closing force to a tissue site. A mechanical means of closing a tissue site may include sutures, staples, hooks, and other devices configured to apply a closing force. Generally, sutures, staples, and other devices may be configured to apply a closing force to a surface of a tissue site or to other tissue peripheral to the tissue site. For example, a thread may be inserted into punctures and drawn across an opening of an incision. The thread may be held under tension with a knot or other securing mechanism to draw opposing sides of an opening together. Sutures and staples may apply a localized stress to tissue near the punctures where the sutures penetrate tissue. Localized stress associated with punctures can lead to patient discomfort, pain, or additional bleeding. In some extreme cases, such mechanical means may cause additional injury, which may result in dehiscence. Weak or delicate skin may be particularity prone to negative effects associated with mechanical means of closure.
(42) These limitations and others may be addressed by the negative-pressure therapy system **100**, which can provide a closing force **131** to facilitate closure of a tissue site. In some embodiments, the negative-pressure therapy system **100** may include a dressing that can be placed over other mechanical closure devices, such as stitches, to provide and distribute a closing force generally perpendicular to a linear tissue site, such as an incision. In some embodiments, the negative-pressure therapy system **100** may apply a closing force that urges opposing sides of an opening in a linear tissue site toward each other, thereby at least partially relieving localized stresses that may be caused by punctures and stitches.

(43) The negative-pressure therapy system **100** may be used on the tissue site **102**. In some embodiments, the contracting layer **114** may be positioned adjacent to the tissue site **102** so that the contracting layer **114** is in contact with the tissue surface **105** surrounding the opening **120**. In some embodiments, the protective layer **116** may be positioned between the contracting layer **114** and the tissue surface **105** surrounding the opening **120**.

(44) In some embodiments, the contracting layer **114** may be a substantially flat or substantially planar body. The contracting layer **114** may have a thickness **126**. In some embodiments, the thickness **126** may be about 15 mm. In other embodiments, the thickness **126** may be thinner or thicker than about 15 mm as needed for the tissue site **102**. In some embodiments, individual portions of the contracting layer **114** may have a minimal tolerance from the thickness **126**. In some embodiments, the thickness **126** may have a tolerance of about 2 mm. The contracting layer **114** may be flexible so that the contracting layer **114** may be contoured to a surface of the tissue site **102**.

(45) In some embodiments, the contracting layer **114** may be formed from thermoplastic elastomers (TPE), such as styrene ethylene butylene styrene (SEBS) copolymers, or thermoplastic polyurethane (TPU). The contracting layer **114** may be formed by combining sheets of TPE or TPU having a thickness between about 0.2 mm and about 2.0 mm to form a structure having the thickness **126**. In some embodiments, the sheets of TPE or TPU may be bonded, welded, adhered, or otherwise coupled to one another. For example, in some embodiments, the sheets of TPE or TPU may be welded using radiant heat, radio-frequency welding, or laser welding. Supracor, Inc., Hexacor, Ltd., Hexcel Corp., and Econocorp, Inc. may produce suitable TPE or TPU sheets for the formation of the contracting layer **114**. In some embodiments, the contracting layer **114** may be formed from a 3D textile, also referred to as a spacer fabric. Suitable 3D textiles may be produced by Heathcoat Fabrics, Ltd., Baltex, and Mueller Textil Group.

(46) In some embodiments, the contracting layer **114** may be formed from a foam. For example, cellular foam, open-cell foam, reticulated foam, or porous tissue collections, may be used to form the contracting layer **114**. In some embodiments, the contracting layer **114** may be formed of GranuFoam®, grey foam, or Zotefoam. Grey foam may be a polyester polyurethane foam having about 60 pores per inch (ppi). Zotefoam may be a closed-cell crosslinked polyolefin foam. In one non-limiting example, the contracting layer **114** may be an open-cell, reticulated polyurethane foam such as GranuFoam® dressing available from Kinetic Concepts, Inc. of San Antonio, Texas; in other embodiments, the contracting layer **114** may be an open-cell, reticulated polyurethane foam such as a VeraFlo® foam, also available from Kinetic Concepts, Inc., of San Antonio, Texas.

(47) In some embodiments, the contracting layer **114** may be formed from a foam that is mechanically or chemically compressed to increase the density of the foam at ambient pressure. A foam that is mechanically or chemically compressed may be referred to as a compressed foam. A compressed foam may be characterized by a firmness factor (FF) that may be defined as a ratio of the density of a foam in a compressed state to the density of the same foam in an uncompressed state. For example, a firmness factor (FF) of 5 may refer to a compressed foam having a density that is five times greater than a density of the same foam in an uncompressed state. Mechanically or chemically compressing a foam may reduce a thickness of the foam at ambient pressure when compared to the same foam that has not been compressed. Reducing a thickness of a foam by mechanical or chemical compression may increase a density of the foam, which may increase the firmness factor (FF) of the foam. Increasing the firmness factor (FF) of a foam may increase a stiffness of the foam in a direction that is parallel to a thickness of the foam. For example, increasing a firmness factor (FF) of the contracting layer **114** may increase a stiffness of the contracting layer **114** in a direction that is parallel to the thickness **126** of the contracting layer **114**. In some embodiments, a compressed foam may be a compressed GranuFoam®. GranuFoam® may have a density of about 0.03 grams per centimeter.sup.3 (g/cm.sup.3) in its uncompressed state. If the GranuFoam® is compressed to have a firmness factor (FF) of 5, the GranuFoam® may be

compressed until the density of the GranuFoam® is about 0.15 g/cm.sup.3. VeraFlo® foam may also be compressed to form a compressed foam having a firmness factor (FF) up to 5.
(48) The firmness factor (FF) may also be used to compare compressed foam materials with non-foam materials. For example, a Supracor® material may have a firmness factor (FF) that allows Supracor® to be compared to compressed foams. In some embodiments, the firmness factor (FF) for a non-foam material may represent that the non-foam material has a stiffness that is equivalent to a stiffness of a compressed foam having the same firmness factor. For example, if a contracting layer is formed from Supracor®, as illustrated in Table 1 below, the contracting layer may have a stiffness that is about the same as the stiffness of a compressed GranuFoam® material having a firmness factor (FF) of 3.
(49) Generally, if a compressed foam is subjected to negative pressure, the compressed foam exhibits less deformation than a similar uncompressed foam. If the contracting layer **114** is formed of a compressed foam, the thickness **126** of the contracting layer **114** may deform less than if the contracting layer **114** is formed of a comparable uncompressed foam. The decrease in deformation may be caused by the increased stiffness as reflected by the firmness factor (FF). If subjected to the stress of negative pressure, the contracting layer **114** formed of compressed foam may flatten less than the contracting layer **114** that is formed from uncompressed foam. Consequently, when negative pressure is applied to the contracting layer **114**, the stiffness of the contracting layer **114** in the direction parallel to the thickness **126** of the contracting layer **114** allows the contracting layer **114** to be more compliant or compressible in other directions, e.g., a direction parallel to the tissue surface **105** or in a direction perpendicular to the opening **120** of the tissue site **102**. The foam material used to form a compressed foam may be either hydrophobic or hydrophilic. The pore size of a foam material may vary according to needs of the contracting layer **114** and the amount of compression of the foam. For example, in some embodiments, an uncompressed foam may have pore sizes in a range of about 400 microns to about 600 microns. If the same foam is compressed, the pore sizes may be smaller than when the foam is in its uncompressed state.
(50) The protective layer **116** may be a layer of material positioned between the contracting layer **114** and the tissue site **102**. In some embodiments, the protective layer **116** may be coextensive with the contracting layer **114**. In other embodiments, the protective layer **116** may be larger or smaller than the contracting layer **114**. In some embodiments, the protective layer **116** may have a thickness that is less than the thickness **126** of the contracting layer **114**. In some embodiments, the protective layer **116** may be a protective mesh, a perforated film, a woven material or a non-woven material. In some embodiments, the protective layer **116** may be laminated to the contracting layer **114**. In some embodiments, the protective layer **116** may inhibit irritation of the tissue site **102**.
(51) FIG. **2** is a plan view, illustrating additional details that may be associated with some embodiments of the contracting layer **114**. The contracting layer **114** may include a plurality of holes **128** or perforations extending through the contracting layer **114** to form walls **130** extending through the contracting layer **114**. In some embodiments, the walls **130** may be generally parallel to the thickness **126** of the contracting layer **114**. In other embodiments, the walls **130** may be generally perpendicular to the surface of the contracting layer **114**. In some embodiments, the holes **128** may have a hexagonal shape as shown.
(52) The contracting layer **114** may cover the opening **120** in the tissue surface **105** of the tissue site **102**. In some embodiments, the contracting layer **114** may have a first orientation line **127** and a second orientation line **129** that is perpendicular to the first orientation line **127**. In some embodiments, an orientation line, such as the first orientation line **127** or the second orientation line **129**, may be a line of symmetry of the contracting layer **114**. A line of symmetry may be, for example, an imaginary line across a surface of the contracting layer **114** defining a fold line such that if the contracting layer **114** is folded on the line of symmetry, the holes **128** and the walls **130** would be coincidentally aligned. Generally, if the dressing **104** is being used, the first orientation line **127** and the second orientation line **129** may be lines used to orient the contracting layer **114**

relative to the tissue site **102**. In some embodiments, the first orientation line **127** and the second orientation line **129** may be used to refer to the desired directions of contraction for the contracting layer **114**. For example, if the first orientation line **127** is oriented parallel to the opening **120**, the desired direction of contraction may be parallel to the second orientation line **129** and perpendicular to the first orientation line **127**. Generally, the contracting layer **114** may be placed at the tissue site **102** so that the first orientation line **127** is parallel to the opening **120** and may cover portions of the tissue surface **105** on both sides of the opening **120**. In some embodiments, the first orientation line **127** may be coincident with the opening **120**.

(53) Although the contracting layer **114** is shown as having a generally rectangular shape including longitudinal edges **132** and latitudinal edges **134**, the contracting layer **114** may have other shapes. For example, the contracting layer **114** may have a diamond, square, or circular shape. In some embodiments, the shape of the contracting layer **114** may be selected to accommodate the type of tissue site being treated. In some embodiments, the first orientation line **127** may be parallel to the longitudinal edges **132**.

(54) Referring more specifically to FIG. **3**, a single hole **128** having a hexagonal shape is shown. The hole **128** may include a center **136** and a perimeter **138**. The hole **128** may have a perforation shape factor (PSF). The perforation shape factor (PSF) may represent an orientation of the hole **128** relative to the first orientation line **127** and the second orientation line **129**. Generally, the perforation shape factor (PSF) is a ratio of ½ a maximum length of the hole **128** that is parallel to the second orientation line **129** to ½ a maximum length of the hole **128** that is parallel to the first orientation line **127**. For reference, the hole **128** may have an X-axis **142** extending through the center **136** between opposing vertices of the hexagon and parallel to the first orientation line **127**, and a Y-axis **140** extending through the center **136** between opposing sides of the hexagon and parallel to the second orientation line **129**. The perforation shape factor (PSF) of the hole **128** may be defined as a ratio of a line segment **144** on the Y-axis **140** extending from the center **136** to the perimeter **138** of the hole **128**, to a line segment **146** on the X-axis **142** extending from the center **136** to the perimeter **138** of the hole **128**. If a length of the line segment **144** is 2.69 mm and the length of the line segment **146** is 2.5 mm, the perforation shape factor (PSF) would be 2.69/2.5 or about 1.08. In other embodiments, the hole **128** may be oriented relative to the first orientation line **127** and the second orientation line **129** so that the perforation shape factor (PSF) may be about 1.07 or 1.1.

(55) Referring to FIG. **4**, a portion of the contracting layer **114** of FIG. **1** is shown. The contracting layer **114** may include the plurality of holes **128** aligned in a pattern of parallel rows. The pattern of parallel rows may include a first row **148** of the holes **128**, a second row **150** of the holes **128**, and a third row **152** of the holes **128**. The centers **136** of the holes **128** in adjacent rows, for example, the first row **148** and the second row **150**, may be characterized by being offset from the second orientation line **129** along the first orientation line **127**. In some embodiments, a line connecting the centers of adjacent rows may form a strut angle (SA) with the first orientation line **127**. For example, a first hole **128**A in the first row **148** may have a center **136**A, and a second hole **128**B in the second row **150** may have a center **136**B. A strut line **154** may connect the center **136**A with the center **136**B. The strut line **154** may form an angle **156** with the first orientation line **127**. The angle **156** may be the strut angle (SA) of the contracting layer **114**. In some embodiments, the strut angle (SA) may be less than about 90°. In other embodiments, the strut angle (SA) may be between about 30° and about 70° relative to the first orientation line **127**. In other embodiments, the strut angle (SA) may be about 66° from the first orientation line **127**. Generally, as the strut angle (SA) decreases, a stiffness of the contracting layer **114** in a direction parallel to the first orientation line **127** may increase. Increasing the stiffness of the contracting layer **114** parallel to the first orientation line **127** may increase the compressibility of the contracting layer **114** perpendicular to the first orientation line **127**. Consequently, if negative pressure is applied to the contracting layer **114**, the contracting layer **114** may be more compliant or compressible in a direction perpendicular

to the first orientation line **127**. By increasing the compressibility of the contracting layer **114** in a direction perpendicular to the first orientation line **127**, the contracting layer **114** may collapse to apply the closing force **131** to the opening **120** of the tissue site **102**, as described in more detail below.

(56) In some embodiments, the centers **136** of the holes **128** in alternating rows, for example, the center **136**A of the first hole **128**A in the first row **148** and a center **136**C of a hole **128**C in the third row **152**, may be spaced from each other parallel to the second orientation line **129** by a length **158**. In some embodiments, the length **158** may be greater than an effective diameter of the hole **128**. If the centers **136** of holes **128** in alternating rows are separated by the length **158**, the walls **130** parallel to the first orientation line **127** may be considered continuous. Generally, the walls **130** may be continuous if the walls **130** do not have any discontinuities or breaks between holes **128**.

(57) Regardless of the shape of the holes **128**, the holes **128** in the contracting layer **114** may leave void spaces in the contracting layer **114** and on the surface of the contracting layer **114** so that only the walls **130** of the contracting layer **114** remain with a surface available to contact the tissue surface **105**. It may be desirable to minimize the walls **130** so that the holes **128** may collapse, causing the contracting layer **114** to collapse and generate the closing force **131** in a direction perpendicular to the first orientation line **127**. However, it may also be desirable not to minimize the walls **130** so much that the contracting layer **114** becomes too fragile for sustaining the application of a negative pressure. The void space percentage (VS) of the holes **128** may be equal to the percentage of the volume or surface area of the void spaces created by the holes **128** to the total volume or surface area of the contracting layer **114**. In some embodiments, the void space percentage (VS) may be between about 40% and about 60%. In other embodiments, the void space percentage (VS) may be about 55%.

(58) In some embodiments, the holes **128** may be formed during molding of the contracting layer **114**. In other embodiments, the holes **128** may be formed by cutting, melting, or vaporizing the contracting layer **114** after the contracting layer **114** is formed. For example, the holes **128** may be formed in the contracting layer **114** by laser cutting the compressed foam of the contracting layer **114**. In some embodiments, an effective diameter of the holes **128** may be selected to permit flow of particulates through the holes **128**. An effective diameter of a non-circular area may be defined as a diameter of a circular area having the same surface area as the non-circular area. In some embodiments, each hole **128** may have an effective diameter of about 3.5 mm. In other embodiments, each hole **128** may have an effective diameter between about 5 mm and about 20 mm. The effective diameter of the holes **128** should be distinguished from the porosity of the material forming the walls **130** of the contracting layer **114**. Generally, an effective diameter of the holes **128** is an order of magnitude larger than the effective diameter of the pores of a material forming the contracting layer **114**. For example, the effective diameter of the holes **128** may be larger than about 1 mm, while the walls **130** may be formed from GranuFoam® material having a pore size less than about 600 microns. In some embodiments, the pores of the walls **130** may not create openings that extend all the way through the material.

(59) Referring now to both FIGS. **2** and **4**, the holes **128** may form a pattern depending on the geometry of the holes **128** and the alignment of the holes **128** between adjacent and alternating rows in the contracting layer **114** with respect to the first orientation line **127**. If the contracting layer **114** is subjected to negative pressure, the holes **128** of the contracting layer **114** may collapse. In some embodiments the void space percentage (VS), the perforation shape factor (PSF), and the strut angle (SA) may cause the contracting layer **114** to collapse along the second orientation line **129** perpendicular to the first orientation line **127** as shown in more detail in FIG. **5**. If the contracting layer **114** is positioned on the tissue surface **105** of the tissue site **102** so that the first orientation line **127** coincides with the opening **120**, the contracting layer **114** may generate the closing force **131** along the second orientation line **129** such that the tissue surface **105** is contracted in the same direction to facilitate closure of the opening **120** and draw the first wall **122**

to the second wall **124** as shown in more detail in FIG. **5**. The closing force **131** may be optimized by adjusting the factors described above, as set forth in Table 1 below. In some embodiments, the holes **128** may be hexagonal, have a strut angle (SA) of approximately 66°, a void space percentage (VS) of about 55%, a firmness factor (FF) of 5, a perforation shape factor (PSF) of 1.07, and an effective diameter of about 5 mm. If the contracting layer **114** is subjected to a negative pressure of about −125 mm Hg, the closing force **131** asserted by the contracting layer **114** may be about 13.3 N. If the effective diameter of the holes **128** of the contracting layer **114** is increased to 10 mm, the closing force **131** may be decreased to about 7.5 N.

(60) FIG. **6** is a perspective sectional view with a portion shown in elevation, illustrating additional details that may be associated with some embodiments of the negative-pressure therapy system **100**. As shown in FIG. **6**, the contracting layer **114** is in the second position, or contracted position, of FIG. **5**. In operation, negative pressure may be supplied to the sealed therapeutic environment **118** with the negative-pressure source **106**. In response to the supply of negative pressure, the contracting layer **114** may collapse from the position illustrated in FIG. **1** to the position illustrated in FIG. **6**. Generally, the thickness **126** of the contracting layer **114** may remain substantially the same. In some embodiments, negative pressure may be supplied to the sealed therapeutic environment **118** until a pressure in the sealed therapeutic environment **118** is about a therapy pressure. In some embodiments, the sealed therapeutic environment **118** may remain at the therapy pressure for a therapy period. In some embodiments, the therapy period may be a time period that allows for opposing sides of the opening **120** to heal. In some embodiments, the therapy period may be cyclic, having a period in which negative pressure may be applied to the tissue site **102** and a period in which negative pressure may be vented from the tissue site **102**. In other embodiments, the therapy period may be longer or as shorter as needed to supply appropriate negative-pressure therapy to the tissue site **102**.

(61) If the contracting layer **114** is in the second position of FIG. **6**, the contracting layer **114** may exert the closing force **131** parallel to the tissue surface **105** of the tissue site **102** toward the opening **120**. The closing force **131** may urge the first wall **122** and the second wall **124** toward one another. In some embodiments, the closing force **131** may close the opening **120**. The closing force **131** may also relieve localized stresses that may be caused by the stitches **103**, reducing the risk of additional trauma to the tissue site **102**.

(62) FIG. **7** is a plan view, illustrating additional details that may be associated with some embodiments of a contracting layer **214**. The contracting layer **214** may be similar to the contracting layer **114** and operate as described above with respect to FIGS. **1-6**. Similar elements may have similar numbers indexed to **200**. For example, the contracting layer **214** is shown as having a generally rectangular shape including longitudinal edges **232** and latitudinal edges **234**. The contracting layer **214** may have a first orientation line **227** and a second orientation line **229** that is perpendicular to the first orientation line **227**. In some embodiments, the first orientation line **227** and the second orientation line **229** may be used to refer to the desired directions of contraction for the contracting layer **214**. For example, if the first orientation line **227** is oriented parallel to the opening **120**, the desired direction of contraction may be parallel to the second orientation line **229** and perpendicular to the first orientation line **227**. Generally, the contracting layer **214** may be placed at the tissue site **102** so that the first orientation line **227** is parallel to the opening **120** and may cover portions of the tissue surface **105** on both sides of the opening **120**. In some embodiments, the first orientation line **227** may be coincident with the opening **120**. The contracting layer **214** may include a plurality of holes **228** or perforations extending through the contracting layer **214**. In some embodiments, the walls **230** of the holes **228** may extend through the contracting layer **214** parallel to the thickness **126** of the contracting layer **214**. In some embodiments, the holes **228** may have a circular shape as shown.

(63) Referring more specifically to FIG. **8**A, a single hole **228** having a circular shape is shown. The hole **228** may include a center **236**, a perimeter **238**, and the perforation shape factor (PSF).

For reference, the hole **228** may have an X-axis **242** extending through the center **236** parallel to the first orientation line **227**, and a Y-axis **240** extending through the center **236** parallel to the second orientation line **229**. In some embodiments, the perforation shape factor (PSF) of the hole **228** may be defined as a ratio of a line segment **244** on the Y-axis **240** extending from the center **236** to the perimeter **238** of the hole **228**, to a line segment **246** on the X-axis **242** extending from the center **236** to the perimeter **238** of the hole **228**. If a length of the line segment **244** is 2.5 mm and the length of the line segment **246** is 2.5 mm, the perforation shape factor (PSF) would be 2.5/2.5 or about 1.

(64) Referring to FIG. **8**B, a portion of the contracting layer **214** of FIG. **7** is shown. The contracting layer **214** may include the plurality of holes **228** aligned in a pattern of parallel rows. The pattern of parallel rows may include a first row **248** of the holes **228**, a second row **250** of the holes **228**, and a third row **252** of the holes **228**. The X-axis **242** of FIG. **8**A of each hole **228** may be parallel to the first orientation line **227** of FIG. **8**B. The centers **236** of the holes **228** in adjacent rows, for example, the first row **248** and the second row **250**, may be characterized by being offset from the second orientation line **229** along the first orientation line **227**. In some embodiments, a line connecting the centers of adjacent rows may form the strut angle (SA) with the first orientation line **227**. For example, a first hole **228**A in the first row **248** may have a center **236**A, and a second hole **228**B in the second row **250** may have a center **236**B. A strut line **254** may connect the center **236**A with the center **236**B. The strut line **254** may form an angle **256** with the first orientation line **227**. The angle **256** may be the strut angle (SA) of the contracting layer **214**. In some embodiments, the strut angle (SA) may be less than about 90°. In other embodiments, the strut angle (SA) may be between about 30° and about 70° relative to the first orientation line **227**. As described above, if negative pressure is applied to the contracting layer **214**, the contracting layer **214** may be more compliant or compressible in a direction perpendicular to the first orientation line **227**. By increasing the compressibility of the contracting layer **214** in a direction perpendicular to the first orientation line **227**, the contracting layer **214** may collapse to apply a closing force to the opening **120** of the tissue site **102**, as described in more detail below.

(65) In some embodiments, the centers **236** of the holes **228** in alternating rows, for example, the center **236**A of the first hole **228**A in the first row **248** and a center **236**C of a hole **228**C in the third row **252**, may be spaced from each other parallel to the second orientation line **229** by a length **258**. In some embodiments, the length **258** may be greater than an effective diameter of the hole **228**. If the centers **236** of holes **228** in alternating rows are separated by the length **258**, the walls **230** parallel to the first orientation line **227** may be considered continuous. Generally, the walls **230** may be continuous if the walls **230** do not have any discontinuities or breaks between holes **228**.

(66) Regardless of the shape of the holes **228**, the holes **228** in the contracting layer **214** may leave void spaces in the contracting layer **214** and on the surface of the contracting layer **214** so that only walls **230** of the contracting layer **214** remain with a surface available to contact the tissue surface **105**. It may be desirable to minimize the walls **230** so that the holes **228** collapse, causing the contracting layer **214** to collapse to generate the closing force **131** in a direction perpendicular to the first orientation line **227**. However, it may also be desirable not to minimize the walls **230** so much that the contracting layer **214** becomes too fragile for sustaining the application of a negative pressure. The void space percentage (VS) of the holes **228** may be equal to the percentage of the volume or surface area of the void spaces created by the holes **228** to the total volume or surface area of the contracting layer **214**. In some embodiments, the void space percentage (VS) may be between about 40% and about 60%. In other embodiments, the void space percentage (VS) may be about 54%.

(67) In some embodiments, a diameter of the holes **228** may be selected to permit flow of particulates through the holes **228**. In some embodiments, each hole **228** may have a diameter of about 5 mm. In other embodiments, each hole **228** may have an effective diameter between about 3.5 mm and about 20 mm.

(68) Referring now to both FIGS. **7** and **8**B, the holes **228** may form a pattern depending on the geometry of the holes **228** and the alignment of the holes **228** between adjacent and alternating rows in the contracting layer **214** with respect to the first orientation line **227**. If the contracting layer **214** is subjected to negative pressure, the holes **228** of the contracting layer **214** may collapse. In some embodiments, the void space percentage (VS), the perforation shape factor (PSF), and the strut angle (SA) may cause the contracting layer **214** to collapse along the second orientation line **229** perpendicular to the first orientation line **227**. If the contracting layer **214** is positioned on the tissue surface **105** of the tissue site **102** so that the first orientation line **227** coincides with the opening **120**, the contracting layer **214** may generate the closing force **131** along the second orientation line **229** such that the tissue surface **105** is contracted in the same direction to facilitate closure of the opening **120**. The closing force **131** may be optimized by adjusting the factors described above as set forth in Table 1 below. In some embodiments, the holes **228** may be circular, have a strut angle (SA) of approximately 37°, a void space percentage (VS) of about 54%, a firmness factor (FF) of 5, a perforation shape factor (PSF) of 1, and a diameter of about 5 mm. If the contracting layer **214** is subjected to a negative pressure of about −125 mm Hg, the contracting layer **214** may assert the closing force **131** of approximately 11.9 N. If the diameter of the holes **228** of the contracting layer **214** is increased to 20 mm, the void space percentage (VS) changed to 52%, the strut angle (SA) changed to 52°, and the perforation shape factor (PSF) and the firmness factor (FF) remain the same, the closing force **131** may be decreased to about 6.5 N.
(69) FIG. **9**A is a plan view, illustrating additional details that may be associated with some embodiments of a contracting layer **314**. The contracting layer **314** may be similar to the contracting layer **114** and operate as described above with respect to FIGS. **1-6**. Similar elements may have similar reference numbers indexed to **300**. The contracting layer **314** may cover the opening **120** in the tissue surface **105** of the tissue site **102**. In some embodiments, the contracting layer **314** may have a first orientation line **327** and a second orientation line **329** that is perpendicular to the first orientation line **327**. In some embodiments, the first orientation line **327** and the second orientation line **329** may be used to refer to the desired directions of contraction for the contracting layer **314**. For example, if the first orientation line **327** is oriented parallel to the opening **120**, the desired direction of contraction may be parallel to the second orientation line **329** and perpendicular to the first orientation line **327**. Generally, the contracting layer **314** may be placed at the tissue site **102** so that the first orientation line **327** is parallel to the opening **120** and may cover portions of the tissue surface **105** on both sides of the opening **120**. In some embodiments, the first orientation line **327** may be coincident with the opening **120**. The contracting layer **314** may include a plurality of holes **328** or perforations extending through the contracting layer **314**. In some embodiments, the walls **330** of the holes **328** may extend through the contracting layer **314** parallel to the thickness **126** of the contracting layer **314**. In some embodiments, the holes **328** may have an ovoid shape as shown.
(70) Referring more specifically to FIG. **10**, a single hole **328** having an ovoid shape is shown. The hole **328** may include a center **336**, a perimeter **338**, and a perforation shape factor (PSF). For reference, the hole **328** may have an X-axis **342** extending through the center **336** parallel to the first orientation line **327**, and a Y-axis **340** extending through the center **336** parallel to the second orientation line **329**. In some embodiments, the perforation shape factor (PSF) of the hole **328** may be defined as a ratio of a line segment **344** on the Y-axis **340** extending from the center **336** to the perimeter **338** of the hole **328**, to a line segment **346** on the X-axis **342** extending from the center **336** to the perimeter **338** of the hole **328**. If a length of the line segment **344** is 2.5 mm and the length of the line segment **346** is 2.5 mm, the perforation shape factor (PSF) would be 2.5/2.5 or about 1.
(71) Referring to FIG. **11**, if the hole **328** is rotated relative to the first orientation line **327** and the second orientation line **329** so that a major axis of the hole **328** is parallel to the second orientation line **329** and a minor axis of the hole **328** is parallel to the first orientation line **327**, the perforation

shape factor (PSF) may change. For example, the perforation shape factor (PSF) is now the ratio of a line segment **360** on the Y-axis **340** extending from the center **336** to the perimeter **338** of the hole **328**, to a line segment **362** on the X-axis **342** extending from the center **336** to the perimeter **338** of the hole **328**. If a length of the line segment **360** is 5 mm and the length of the line segment **362** is 2.5 mm, the perforation shape factor (PSF) would be 5/2.5 or about 2.

(72) Referring to FIG. **12**, if the hole **328** is rotated relative to the first orientation line **327** and the second orientation line **329** so that a major axis of the hole **328** is parallel to the first orientation line **327** and a minor axis of the hole **328** is parallel to the second orientation line **329**, the perforation shape factor (PSF) may change. For example, the perforation shape factor (PSF) is now the ratio of a line segment **364** on the Y-axis **340** extending from the center **336** to the perimeter **338** of the hole **328**, to a line segment **366** on the X-axis **342** extending from the center **336** to the perimeter **338** of the hole **328**. If a length of the line segment **364** is 2.5 mm and the length of the line segment **366** is 5 mm, the perforation shape factor (PSF) would be 2.5/5 or about ½.

(73) Referring to FIG. **9**B, a portion of the contracting layer **314** of FIG. **9**A is shown. The contracting layer **314** may include the plurality of holes **328** aligned in a pattern of parallel rows. The pattern of parallel rows may include a first row **348** of the holes **328**, a second row **350** of the holes **328**, and a third row **352** of the holes **328**. The X-axis **342** of FIGS. **10**, **11**, and **12** of each hole **328** may be parallel to the first orientation line **327** of FIG. **9**B. The centers **336** of the holes **328** in adjacent rows, for example, the first row **348** and the second row **350**, may be characterized by being offset from the second orientation line **329** along the first orientation line **327**. In some embodiments, a line connecting the centers of adjacent rows may form a strut angle (SA) with the first orientation line **327**. For example, a first hole **328**A in the first row **348** may have a center **336**A, and a second hole **328**B in the second row **350** may have a center **336**B. A strut line **354** may connect the center **336**A with the center **336**B. The strut line **354** may form an angle **356** with the first orientation line **327**. The angle **356** may be the strut angle (SA) of the contracting layer **314**. In some embodiments, the strut angle (SA) may be less than about 90°. In other embodiments, the strut angle (SA) may be between about 30° and about 70° relative to the first orientation line **327**. As described above, if negative pressure is applied to the contracting layer **314**, the contracting layer **314** may be more compliant or compressible in a direction perpendicular to the first orientation line **327**. By increasing the compressibility of the contracting layer **314** in a direction perpendicular to the first orientation line **327**, the contracting layer **314** may collapse to apply the closing force **131** to the opening **120** of the tissue site **102**, as described in more detail below.

(74) In some embodiments, the centers **336** of the holes **328** in alternating rows, for example, the center **336**A of the first hole **328**A in the first row **348** and a center **336**C of a hole **328**C in the third row **352**, may be spaced from each other parallel to the second orientation line **329** by a length **358**. In some embodiments, the length **358** may be greater than an effective diameter of the hole **328**. If the centers **336** of holes **328** in alternating rows are separated by the length **358**, the walls **330** parallel to the first orientation line **327** may be considered continuous. Generally, the walls **330** may be continuous if the walls **330** do not have any discontinuities or breaks between holes **328**.

(75) Regardless of the shape of the holes **328**, the holes **328** in the contracting layer **314** may leave void spaces in the contracting layer **314** and on the surface of the contracting layer **314** so that only walls **330** of the contracting layer **314** remain with a surface available to contact the tissue surface **105**. It may be desirable to minimize the walls **330** so that the holes **328** may collapse, causing the contracting layer **314** to collapse the closing force **131** in a direction perpendicular to the first orientation line **327**. However, it may also be desirable not to minimize the walls **330** so much that the contracting layer **314** becomes too fragile for sustaining the application of a negative pressure. The void space percentage (VS) of the holes **328** may be equal to the percentage of the volume or surface area of the void spaces created by the holes **328** to the total volume or surface area of the contracting layer **314**. In some embodiments, the void space percentage (VS) may be between about 40% and about 60%. In other embodiments, the void space percentage (VS) may be about

56%.

(76) In some embodiments, an effective diameter of the holes **328** may be selected to permit flow of particulates through the holes **328**. In some embodiments, each hole **328** may have an effective diameter of about 7 mm. In other embodiments, each hole **328** may have an effective diameter between about 2.5 mm and about 20 mm.

(77) Referring now to both FIGS. **9**A and **9**B, the holes **328** may form a pattern depending on the geometry of the holes **328** and the alignment of the holes **328** between adjacent and alternating rows in the contracting layer **314** with respect to the first orientation line **327**. If the contracting layer **314** is subjected to negative pressure, the holes **328** of the contracting layer **314** may collapse, causing the contracting layer **314** to collapse along the second orientation line **329** perpendicular to the first orientation line **327**. If the contracting layer **314** is positioned on the tissue surface **105** of the tissue site **102** so that the first orientation line **327** coincides with the opening **120**, the contracting layer **314** may generate the closing force **131** along the second orientation line **329** such that the tissue surface **105** is contracted in the same direction to facilitate closure of the opening **120**. The closing force **131** may be optimized by adjusting the factors described above as set forth in Table 1 below. In some embodiments, the holes **328** may be ovular, have a strut angle (SA) of approximately 47°, a void space percentage (VS) of about 56%, a firmness factor (FF) of 5, a perforation shape factor (PSF) of 1, and an effective diameter of about 7 mm (where the major axis is about 10 mm and the minor axis is about 5 mm). If the contracting layer **314** is subjected to a negative pressure of about −125 mm Hg, the contracting layer **314** may assert the closing force **131** of approximately 13.5 N.

(78) FIG. **13**A is a plan view, illustrating additional details that may be associated with some embodiments of a contracting layer **414**. The contracting layer **414** may be similar to the contracting layer **114** and operate as described with respect to FIGS. **1-6**. Similar elements may have similar reference numbers indexed to **400**. For example, the contracting layer **414** is shown as having a generally rectangular shape including longitudinal edges **432** and latitudinal edges **434**. The contracting layer **414** may cover the opening **120** in the tissue surface **105** of the tissue site **102**. In some embodiments, the contracting layer **414** may have a first orientation line **427** and a second orientation line **429** that is perpendicular to the first orientation line **427**. In some embodiments, the first orientation line **427** and the second orientation line **429** may be used to refer to the desired directions of contraction for the contracting layer **414**. For example, if the first orientation line **427** is oriented parallel to the opening **120**, the desired direction of contraction may be parallel to the second orientation line **429** and perpendicular to the first orientation line **427**. Generally, the contracting layer **414** may be placed at the tissue site **102** so that the first orientation line **427** is parallel to the opening **120** and may cover portions of the tissue surface **105** on both sides of the opening **120**. In some embodiments, the first orientation line **427** may be coincident with the opening **120**. The contracting layer **414** may include a plurality of holes **428** or perforations extending through the contracting layer **414**. In some embodiments, the walls **430** of the holes **428** may extend through the contracting layer **414** parallel to the thickness **126** of the contracting layer **414**. In some embodiments, the holes **428** may have a triangular shape as shown.

(79) Referring more specifically to FIG. **14**, a single hole **428** having a triangular shape is shown. The hole **428** may include a center **436**, a perimeter **438**, and a perforation shape factor (PSF). In some embodiments, the hole **428** may include a first vertex **460**, a second vertex **462**, and a third vertex **464**. For reference, the hole **428** may have an X-axis **442** extending through the center **436** parallel to the first orientation line **427**, and a Y-axis **440** extending through the center **436** parallel to the second orientation line **429**. In some embodiments, the perforation shape factor (PSF) of the hole **428** may be defined as a ratio of a line segment **444** on the Y-axis **440** extending from the center **436** to the perimeter **438** of the hole **428**, to a line segment **446** on the X-axis **442** extending from the center **436** to the perimeter **438** of the hole **428**. If a length of the line segment **444** is 1.1 mm and the length of the line segment **446** is 1 mm, the perforation shape factor (PSF) would be

1.1/1 or about 1.1.

(80) Referring to FIG. **13**B, a portion of the contracting layer **414** of FIG. **13**A is shown. The contracting layer **414** may include the plurality of holes **428** aligned in a pattern of parallel rows. The pattern of parallel rows may include a first row **448** of the holes **428**, a second row **450** of the holes **428**, and a third row **452** of the holes **428**. The X-axis **442** of FIG. **14** of each hole **428** may be parallel to the first orientation line **427** of FIG. **13**B. In some embodiments, a first hole **428**A in the first row **448** may be oriented so that the first vertex **460**A of a first hole **428**A may be between the first orientation line **427** and a leg of the first hole **428**A opposite the first vertex **460**A. A hole **428**C that is adjacent the first hole **428**A in the first row **448** may be oriented so that the first vertex **460**C may be oriented opposite the first hole **428**A.

(81) The centers **436** of the holes **428** in adjacent rows having the first vertex **460** oriented in a same direction, for example, the first row **448** and the second row **450**, may be characterized by being offset from the second orientation line **429** along the first orientation line **427**. In some embodiments, a line connecting the centers of adjacent rows may form a strut angle (SA) with the first orientation line **427**. For example, a first hole **428**A in the first row **448** may have a center **436**A, and a second hole **428**B in the second row **450** may have a center **436**B and a first vertex **460**B. A strut line **454** may connect the center **436**A with the center **436**B. The strut line **454** may form an angle **456** with the first orientation line **427**. The angle **456** may be the strut angle (SA) of the contracting layer **414**. In some embodiments, the strut angle (SA) may be less than about 90°. In other embodiments, the strut angle (SA) may be between about 40° and about 70° relative to the first orientation line **427**. As described above, if negative pressure is applied to the contracting layer **414**, the contracting layer **414** may be more compliant or compressible in a direction perpendicular to the first orientation line **427**. By increasing the compressibility of the contracting layer **414** in a direction perpendicular to the first orientation line **427**, the contracting layer **414** may collapse to apply the closing force **131** to the opening **120** of the tissue site **102**, as described in more detail below.

(82) Regardless of the shape of the holes **428**, the holes **428** in the contracting layer **414** may leave void spaces in the contracting layer **414** and on the surface of the contracting layer **414** so that only walls **430** of the contracting layer **414** remain with a surface available to contact the tissue surface **105**. It may be desirable to minimize the walls **430** so that the holes **428** may collapse, causing the contracting layer **414** to generate the closing force **131** in a direction perpendicular to the first orientation line **427**. However, it may also be desirable not to minimize the walls **430** so much that the contracting layer **414** becomes too fragile for sustaining the application of a negative pressure. The void space percentage (VS) of the holes **428** may be equal to the percentage of the volume or surface area of the void spaces created by the holes **428** to the total volume or surface area of the contracting layer **414**. In some embodiments, the void space percentage (VS) may be between about 40% and about 60%. In other embodiments, the void space percentage (VS) may be about 56%.

(83) In some embodiments, an effective diameter of the holes **428** may be selected to permit flow of particulates through the holes **428**. In some embodiments, each hole **428** may have an effective diameter of about 7 mm. In other embodiments, each hole **428** may have an effective diameter between about 2.5 mm and about 20 mm.

(84) Referring now to both FIGS. **13**A and **13**B, the holes **428** may form a pattern depending on the geometry of the holes **428** and the alignment of the holes **428** between adjacent and alternating rows in the contracting layer **414** with respect to the first orientation line **427**. If the contracting layer **414** is subjected to negative pressure, the holes **428** of the contracting layer **414** may collapse. In some embodiments, the void space percentage (VS), the perforation shape factor (PSF), and the strut angle (SA) may cause the contracting layer **414** to collapse along the second orientation line **429** perpendicular to the first orientation line **427**. If the contracting layer **414** is positioned on the tissue surface **105** of the tissue site **102** so that the first orientation line **427** coincides with the

opening **120**, the contracting layer **414** may generate the closing force **131** along the second orientation line **429** such that the tissue surface **105** is contracted in the same direction to facilitate closure of the opening **120**. The closing force **131** may be optimized by adjusting the factors described above as set forth in Table 1 below. In some embodiments, the holes **428** may be triangular, have a strut angle (SA) of approximately 63°, a void space percentage (VS) of about 40%, a firmness factor (FF) of 5, a perforation shape factor (PSF) of 1.1, and an effective diameter of about 10 mm. If the contracting layer **414** is subjected to a negative pressure of about −125 mm Hg, the contracting layer **414** may assert the closing force **131** of approximately 12.2 N.
(85) FIG. **15** is a plan view, illustrating additional details that may be associated with some embodiments of a contracting layer **514**. The contracting layer **514** may be similar to the contracting layer **114** described above with respect to FIGS. **1-6**. The contracting layer **514** may include stripes **516** having a first density and stripes **518** having a second density. In some embodiments, the stripes **516** and the stripes **518** may be vertically oriented relative to a tissue site, and in other embodiments, the stripes **516** and the stripes **518** may be horizontally oriented relative to a tissue site. In still other embodiments, the stripes **516** and the stripes **518** may be oriented at an angle relative to a tissue site. In some embodiments, the second density may be greater than the first density. In some embodiments, the second density may be between about 3 times and about 5 times greater than the first density. In some embodiments, the contracting layer **514** may be formed from a foam, similar to GranuFoam®. In some embodiments, the stripes **518** may be formed by compressing portions of the foam. For example, the stripes **516** may be an uncompressed foam, and the stripes **518** may be a compressed foam having a firmness factor of about 5. Generally, the stripes **516** may be more compressible than the stripes **518**. If the contracting layer **514** is placed under a negative-pressure, the stripes **516** may collapse before the stripes **518**. In some embodiments, if the stripes **516** collapse, the contracting layer **514** may compress perpendicular to the stripes **516**.
(86) A closing force, such as the closing force **131**, generated by a contracting layer, such as the contracting layer **114**, may be related to a compressive force generated by applying negative pressure at a therapy pressure to a sealed therapeutic environment. For example, the closing force **131** may be proportional to a product of a therapy pressure (TP) in the sealed therapeutic environment **118**, the compressibility factor (CF) of the contracting layer **114**, and a surface area (A) of the contracting layer **114**. The relationship is expressed as follows:
Closing force α(TP*CF*A)
(87) In some embodiments, the therapy pressure TP is measured in N/m.sup.2, the compressibility factor (CF) is dimensionless, the area (A) is measured in m.sup.2, and the closing force is measured in Newtons (N). The compressibility factor (CF) resulting from the application of negative pressure to a contracting layer may be, for example, a dimensionless number that is proportional to the product of the void space percentage (VS) of a contracting layer, the firmness factor (FF) of the contracting layer, the strut angle (SA) of the holes in the contracting layer, and the perforation shape factor (PSF) of the holes in the contracting layer. The relationship is expressed as follows:
Compressibility Factor (CF)α(VS*FF*sin(SA)*PSF)
(88) Based on the above formulas, contracting layers formed from different materials with holes of different shapes were manufactured and tested to determine the closing force of the contracting layers. For each contracting layer, the therapy pressure TP was about −125 mmHg and the dimensions of the contracting layer were about 200 mm by about 53 mm so that the surface area (A) of the contracting layer was about 106 cm.sup.2 or 0.0106 m.sup.2. Based on the two equations described above, the closing force for a Supracor® contracting layer **114** having a firmness factor (FF) of 3 was about 13.3 where the Supracor® contracting layer **114** had hexagonal holes **128** with a distance between opposite vertices of 5 mm, a perforation shape factor (PSF) of 1.07, a strut angle (SA) of approximately 66°, and a void space percentage (VS) of about 55%. A similarly dimensioned GranuFoam® contracting layer **114** generated the closing force **131** of about 9.1

Newtons (N).
(89) TABLE-US-00001 TABLE 1 Material VS FF SA Hole Shape PSF Major diam. (mm) Closing force GranuFoam ® 56 5 47 Ovular 1 10 13.5 Supracor ® 55 3 66 Hexagon 1.1 5 13.3 GranuFoam ® 40 5 63 Triangle 1.1 10 12.2 GranuFoam ® 54 5 37 Circular 1 5 11.9 GranuFoam ® 52 5 37 Circular 1 20 10.3 Grey Foam N/A 5 N/A Horizontal stripes N/A N/A 9.2 GranuFoam ® 55 5 66 Hexagon 1.1 5 9.1 GranuFoam ® N/A 5 N/A Horizontal stripes N/A N/A 8.8 Zotefoam 52 3 37 Circular 1 10 8.4 GranuFoam ® 52 5 37 Circular 1 10 8.0 GranuFoam ® 52 5 64 Circular 1 10 7.7 GranuFoam ® 56 5 66 Hexagon 1.1 10 7.5 Grey Foam N/A 3 N/A Horizontal stripes N/A N/A 7.2 Zotefoam 52 3 52 Circular 1 20 6.8 GranuFoam ® N/A 3 N/A Horizontal Striping N/A N/A 6.6 GranuFoam ® 52 5 52 Circular 1 20 6.5 GranuFoam ® N/A 5 N/A Vertical Stripes N/A N/A 6.1 GranuFoam ® N/A 1 N/A None N/A N/A 5.9 GranuFoam ® N/A 3 N/A Vertical stripes N/A N/A 5.6 GranuFoam ® 52 1 37 None 1 10 5.5
(90) In some embodiments, the formulas described above may not precisely describe the closing forces due to losses in force due to the transfer of the force from the contracting layer to the wound. For example, the modulus and stretching of the cover **112**, the modulus of the tissue site **102**, slippage of the cover **112** over the tissue site **102**, and friction between the protective layer **116**, the contracting layer **114**, and the tissue surface **105** of tissue site **102** may cause the actual value of the closing force **131** to be less than the calculated value of the closing force **131**.
(91) In some embodiments, the material, the void space percentage (VS), the firmness factor, the strut angle, the hole shape, the perforation shape factor (PSF), and the hole diameter may be selected to increase compression or collapse of the contracting layer **114** in a lateral direction, as shown by the closing force **131**, by forming weaker walls **130**. Conversely, the factors may be selected to decrease compression or collapse of the contracting layer **114** in a lateral direction, as shown by the closing force **131**, by forming stronger walls **130**. Similarly, the factors described herein can be selected to decrease or increase the compression or collapse of the contracting layer **114** perpendicular to the closing force **131**.
(92) The systems, apparatuses, and methods described herein may provide significant advantages. For example, closing forces generated by the described contracting layers meet or exceed other contracting layers designed for a similar purpose. The described contracting layers may also assist in closure of an incisional tissue site by distributed force along a length of the incisional opening, reducing potential trauma that may be caused by point loading such as with sutures, staples or hooks.
(93) While shown in a few illustrative embodiments, a person having ordinary skill in the art will recognized that the systems, apparatuses, and methods described herein are susceptible to various changes and modifications. Moreover, descriptions of various alternatives using terms such as “or” do not require mutual exclusivity unless clearly required by the context, and the indefinite articles “a” or “an” do not limit the subject to a single instance unless clearly required by the context.
(94) The appended claims set forth novel and inventive aspects of the subject matter described above, but the claims may also encompass additional subject matter not specifically recited in detail. For example, certain features, elements, or aspects may be omitted from the claims if not necessary to distinguish the novel and inventive features from what is already known to a person having ordinary skill in the art. Features, elements, and aspects described herein may also be combined or replaced by alternative features serving the same, equivalent, or similar purpose without departing from the scope of the invention defined by the appended claims.

Claims

1. A dressing for closing an opening through a surface of a tissue site, the dressing comprising: a tissue interface, comprising: a contracting layer comprising a porous foam and a plurality of holes extending through the contracting layer to form a void space distinct from pores within the porous

foam, wherein the contracting layer further comprises a higher stiffness in a first direction parallel to a thickness of the contracting layer than a second direction perpendicular the thickness, and a protective layer configured to be positioned between the contracting layer and at least a portion of the surface of the tissue site surrounding the opening; and a sealing drape configured to cover the tissue interface to form a sealed space, wherein the protective layer extends beyond the contracting layer into contact with a portion of the sealing drape.

2. The dressing of claim 1, wherein the plurality of holes have a perforation shape factor and a strut angle configured to cause the plurality of holes to collapse in a direction substantially perpendicular to the opening, wherein the contracting layer generates a closing force substantially parallel to the surface of the tissue site to close the opening in response to an application of negative pressure, wherein the strut angle is an angle between a line connecting centers of holes in adjacent rows and an orientation line perpendicular to a direction of the closing force, and wherein the perforation shape factor of a hole of the plurality of holes is a ratio of a first line segment to a second line segment, the first line segment extending from a center of the hole to a perimeter of the hole in a direction perpendicular to a direction of the closing force and the second line segment extending from a center of the hole to a perimeter of the hole in a direction parallel to the direction of the closing force.

3. The dressing of claim 2, wherein the strut angle is about 90 degrees.

4. The dressing of claim 2, wherein the strut angle is less than about 90 degrees.

5. The dressing of claim 2, wherein the perforation shape factor of the hole is less than about 1.

6. The dressing of claim 2, wherein the perforation shape factor of the hole is more than about 1.

7. The dressing of claim 1, wherein the plurality of holes are formed in two or more parallel rows, and wherein the plurality of holes in a first row are offset from the plurality of holes in a second row adjacent to the first row.

8. The dressing of claim 7, wherein the plurality of holes have an average effective diameter, and wherein a first center of one of the holes in the first row is spaced apart from a second center of one of the holes in the second row by a length greater than the effective diameter.

9. The dressing of claim 1, wherein the contracting layer is formed from a foam material having a firmness factor, wherein the firmness factor is a ratio of density of the foam material in a compressed state to a density of the foam material in an uncompressed state.

10. The dressing of claim 9, wherein the firmness factor is about 5.

11. The dressing of claim 9, wherein the firmness factor is about 3.

12. The dressing of claim 1, wherein a shape of each hole of the plurality of holes is hexagonal.

13. The dressing of claim 1, wherein a shape of each hole of the plurality of holes is elliptical.

14. The dressing of claim 1, wherein a shape of each hole of the plurality of holes is circular.

15. The dressing of claim 1, wherein the contracting layer comprises a compressed foam.

16. The dressing of claim 1, wherein the contracting layer comprises a felted foam.

17. The dressing of claim 1, wherein the contracting layer comprises a 3D spacer fabric.

18. The dressing of claim 1, wherein the contracting layer comprises a thermoplastic elastomer.

19. The dressing of claim 1, wherein the contracting layer comprises a thermoplastic polyurethane.

20. The dressing of claim 1, wherein the contracting layer has a thickness of about 15 mm.

21. The dressing of claim 1, wherein the protective layer comprises a mesh.

22. The dressing of claim 1, wherein the protective layer comprises a perforated film.

23. The dressing of claim 1, wherein the protective layer is configured to inhibit irritation of the tissue site.

24. The dressing of claim 1, wherein the protective layer is laminated to the contracting layer.

25. The dressing of claim 1, wherein the protective layer is configured to be positioned over the opening.